

SUSTAINABLE TRANSPORTATION

Why is This Issue Important?

1. Regular transportation causes air pollution
2. There are limited fossil fuels
3. Wasted resources on a micro scale



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Air Pollution



- #1 pollutant in oil driven vehicles is CO_2 , or carbon dioxide .

Air Pollution & Your Health

- Between 350,000 and 500,000 Chinese die prematurely each year because of the country's disastrous air pollution



Beijing, China

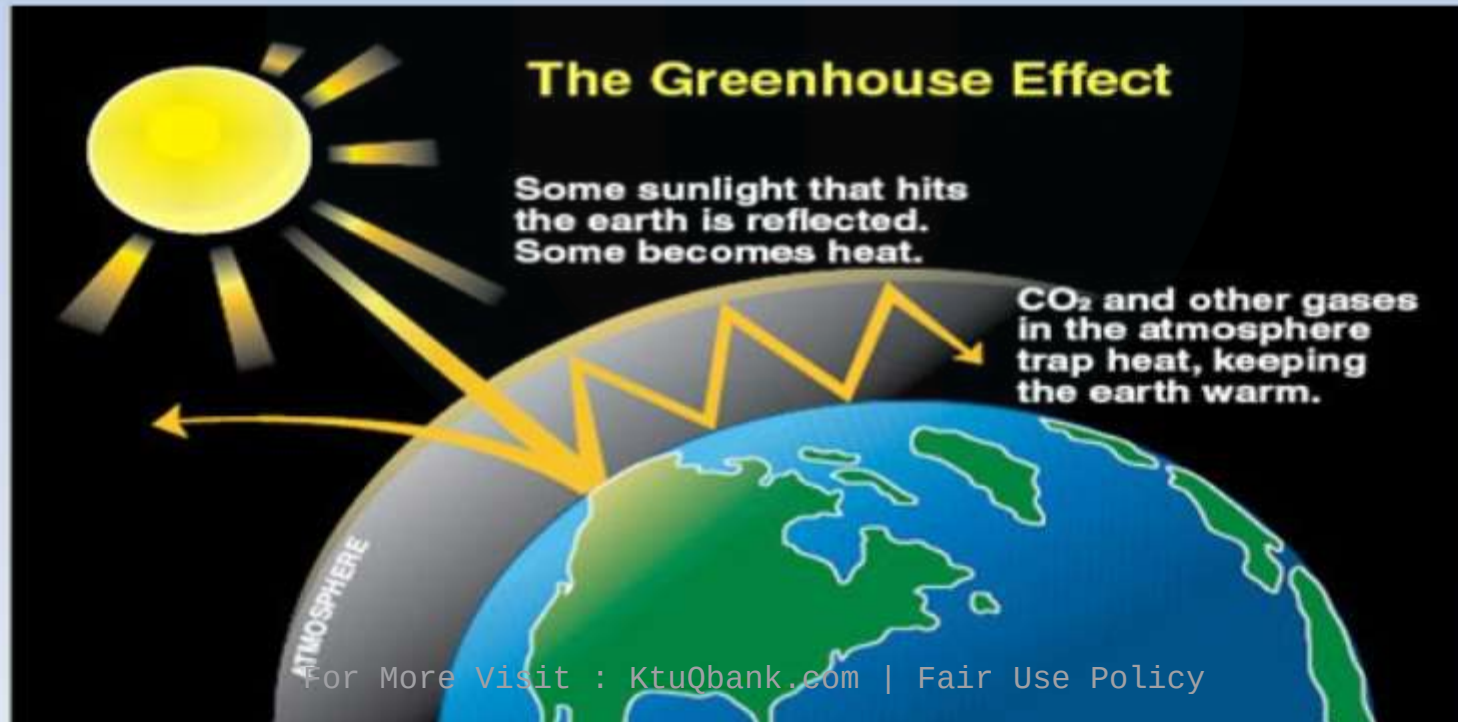
Air Pollution & Transportation

- This pollution comes partially from a vehicle's **exhaust**, which outputs large amounts of CO_2 .



Air Pollution & The Environment

- A **greenhouse gas** is any gaseous compound in the atmosphere that is capable of absorbing infrared radiation, thereby trapping and holding heat in the atmosphere. Increasing the heat in the atmosphere ultimately leads to global warming.
- In 2012, U.S. greenhouse gas emissions totaled 6,526 million metric tons of carbon dioxide equivalents, which equaled 82% of all human caused greenhouse gasses.



Limited Fossil Fuels

- “**Fossil fuel**” is a broad term that refers to combustible (explosive) deposits of organic materials found on Earth.
- These fuels are formed by decayed plants and animals that naturally convert into **crude oil, coal, and natural gas** over hundreds of millions of years. This extreme creation duration makes the current supply finite (for human intents and purposes).
- Eventually, we will either run out of fossil fuels or go extinct as a species. We *must* find renewable energy sources before it gets to this end.



Wasted Resources in YOUR Life!

- The average American commuter spends around 38 hours in traffic per calendar year.
- Vehicles continue to idle in traffic, outputting CO₂ and consuming fuel that is simply **wasteful.**



What is Sustainable Transportation?

- **Sustainable transportation** is human transportation that does not rely on fossil fuels or other finite resources, does not harm the environment, and still allows mankind to travel the world as quickly and safely as they do today.

Potential Solutions

- Many solutions are being implemented *today* across the globe in order to overcome these challenges.
- While these solutions are a slow start, they are a step in the right direction to achieve full sustainability in our transportation systems.



Air Pollution Solutions

- Lowering or eliminating the use of oil
- Fully electric vehicles
- Hybrid vehicles



Air Pollution Solutions

- **Cleaner energy production** will decrease air pollution dramatically.
- Cleaner energy alternatives to oil include **hydroelectric, solar, and nuclear**.



Solutions Expanded

- These proposed solutions are for the American public and would work practically *on a global scale*.
- The most efficient transportation systems aren't individual vehicles, it's **mass transportation**. This includes bus systems, high speed trains, light rails, trolleys, etc.
- Mass transportation systems allows many people to utilize the system simultaneously, while using much less energy and emitting far less per person.
- These systems are perfect for compressed areas such as cities, but not yet viable for *all* areas.



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What Can YOU Do Today?

- Consider **voting** for candidates willing to make governmental mandates to improve sustainable transportation options, such as green bus systems or electric trains.
- Consider utilizing **mass transportation** more often as a substitute for your personal vehicle.
- When you do drive, you should also **consider changing your driving style**. Why is this important? Let's take a look...

Manufacturer Strides

- Teaching people to drive more economically



More Long Term Goals for Our Society

- We need to **improve the convenience of using an electric vehicle** .
- We need to **make mass transit more viable** in more parts of the country.
- Most importantly, we need to **change how we perceive transportation and its effect on our earth as a society** .



The Ideal World



- Transportation that requires zero fossil fuels.
- Electric personal vehicles, electric mass transportation, and this electricity being harvested from clean sources (hydroelectric, solar, nuclear, etc.)
- Transportation that is safe, fast, and convenient for all citizens, regardless of their destination.

Features of Sustainable Transportation

- Needed of good planning for sustainability. **On the roads, priority must be given to pedestrians, cyclists, public transport usage, specialist service vehicles for emergency and waste disposal and other motor traffic in that order.**
- The streets also must be provided with car free zones giving priority to pedestrians and non- motorized vehicles. Cars must be used only when there is sufficient number of travelers. Public transport system should provide maximum connectivity to schools and residential areas. Other motor vehicles must be limited to designated roads.
- The vehicles which use alternatives energy sources are to be promoted for the sake of environment. This will reduce GHG emissions considerably.
- Congestion on roads can be minimized by promoting the use of rails and other public transport systems

- The public transport system must be efficient. There should be well-designed vehicles which use hybrid energy principles. The public transport vehicles should be maintained properly. Good personnel management is also essential for providing better services. Metro rail and mono rail projects in cities will reduce the congestion
- On the roads there must be facilities like bus bays, bus lines and special priority for public transport vehicles at signals.
- Walking and cycling for short distances are to be made attractive by providing prioritized paths in streets and cities.
- Walking and cycling will reduce the GCG emissions. Moreover it is healthy as well. Subways must be provided for the safety of pedestrians.

- Taxes and charges must be imposed on private motor vehicles in order to promote public transport system
- Routing and scheduling of the public transport system are to be modified in accordance with public opinion. There must be easy access to the public by providing online facilities for ticket booking and awareness about the time schedules and routes. Modern communication systems are also to be used for assisting the public to access the transport facilities.
- Offices and other institutions must be located as close to each other as possible and thus movements of people can be optimized.

GREEN BUILDING



INTRODUCTION

A Green Building is one which uses less water, optimizes energy efficiency, conserves natural resources, generates less waste and provides healthier space for occupants as compared to conventional buildings.

OBJECTIVES OF GREEN BUILDING

- Green Buildings are designed to reduce the overall impact on human health and the natural environment by the following ways:
- Using energy, water and other resources efficiently.
- By reducing waste, pollution, and environmental degradation.

FUNDAMENTAL PRINCIPLES

- Structure design efficiency
- Energy efficiency
- Water efficiency
- Reuse of Materials
- Waste and toxic reduction



- **STRUCTURE EFFICIENCY:**

- It is the concept of sustainable building and has largest impact on cost and performance.
- It aims to minimize the environment impact associated with all life-cycles.



- **ENERGY EFFICIENCY:**

- The layout of the construction can be strategised so that natural light enters the building.
- Shading the roof with trees offers an eco-friendly alternative to air conditioning.

- **WATER EFFICIENCY:**

- To minimize water consumption one should aim to use the water which has been collected, used, purified and reused.



REUSE OF MATERIALS

- It is probable to reuse resources.
- What may be waste to us may have another benefit to something else.



INDIAN GREEN BUILDING COUNCIL

- The Indian Green Building Council (IGBC) was formed in the year 2001 by Confederation of Indian Industry (CII).
- The aim of the council is to bring green building movement in India and facilitate India to become one of the global leaders in green buildings.



IGBC RATING SYSTEM

- IGBC has developed green building rating programmes to cover commercial, residential, factory buildings, etc.
- Each rating system divided into different levels of certification are as follows:
 - "Certified" to recognise best practices.
 - "Silver" to recognise outstanding performance.
 - "Gold" to recognise national excellence.
 - "Platinum" to recognise global leadership.

Methods to make a building energy efficient

- **Passive solar energy utilization**
- **Thermal storage**
- **Cooling strategies**
- **Reducing electricity usage**
- **High performance insulation**

GREEN BUILDINGS PROJECT IN INDIA

- Suzlon Energy Limited-Pune
- Biodiversity Conservation India-Bangalore
- Olympia Technology Park-Chennai
- ITC Green Centre-Gurgaon
- The Druk White Lotus School-Ladakh
- Doon School-Dehradun
- Raintree Hotels-Chennai
- Nokia-Gurgaon
- Rajiv Gandhi International Airport-Hyderabad
- Hiranandini-BG House, Powai
- ABN Amro Bank, Chennai
- Palais Royale at Worli, Mumbai
- Punjab Forest Complex, Mohali



- **SUZLON ENERGY LIMITED,PUNE:**
 - Several accolades continue to shower upon Suzlon's global headquarter in Pune "One Earth".
 - LEED certified it as „PLATINUM“ and it is built on an area of 10.13 acres.
 - One Earth can be counted as among the largest green building projects in India.





- **RAJIV GANDHI INTERNATIONAL AIRPORT- HYDERABAD:**
 - India's first Greenfield airport is undeniably among the top 10 green buildings in India.
 - First airport in asia to be certified with „SILVER“ rating.
 - This green building ensures optimal use of natural light and minimal wastage of electricity or energy consumption.



- **NOKIA-GURGAON:**

- „GOLD” rated building by LEED.



SOME IMAGES OF GREEN BUILDINGS



DIFFERENT FROM OTHER BUILDINGS

- The design, maintaince and construction of buildings have tremendous effect on our enviornment and natural resources.
- Green Building is different from the other buildings because it use a **minimum amount of nonrenewable energy, produce minimal pollution, increases the comfort, health and safety of the people** who work in them.
- It also minimize the **waste in construction by recovering materials and reusing or recycling them.**

ENVIRONMENTAL BENEFITS

- Protect biodiversity and ecosystems
- Improve air and water quality
- Reduce waste streams
- Conserve natural resources

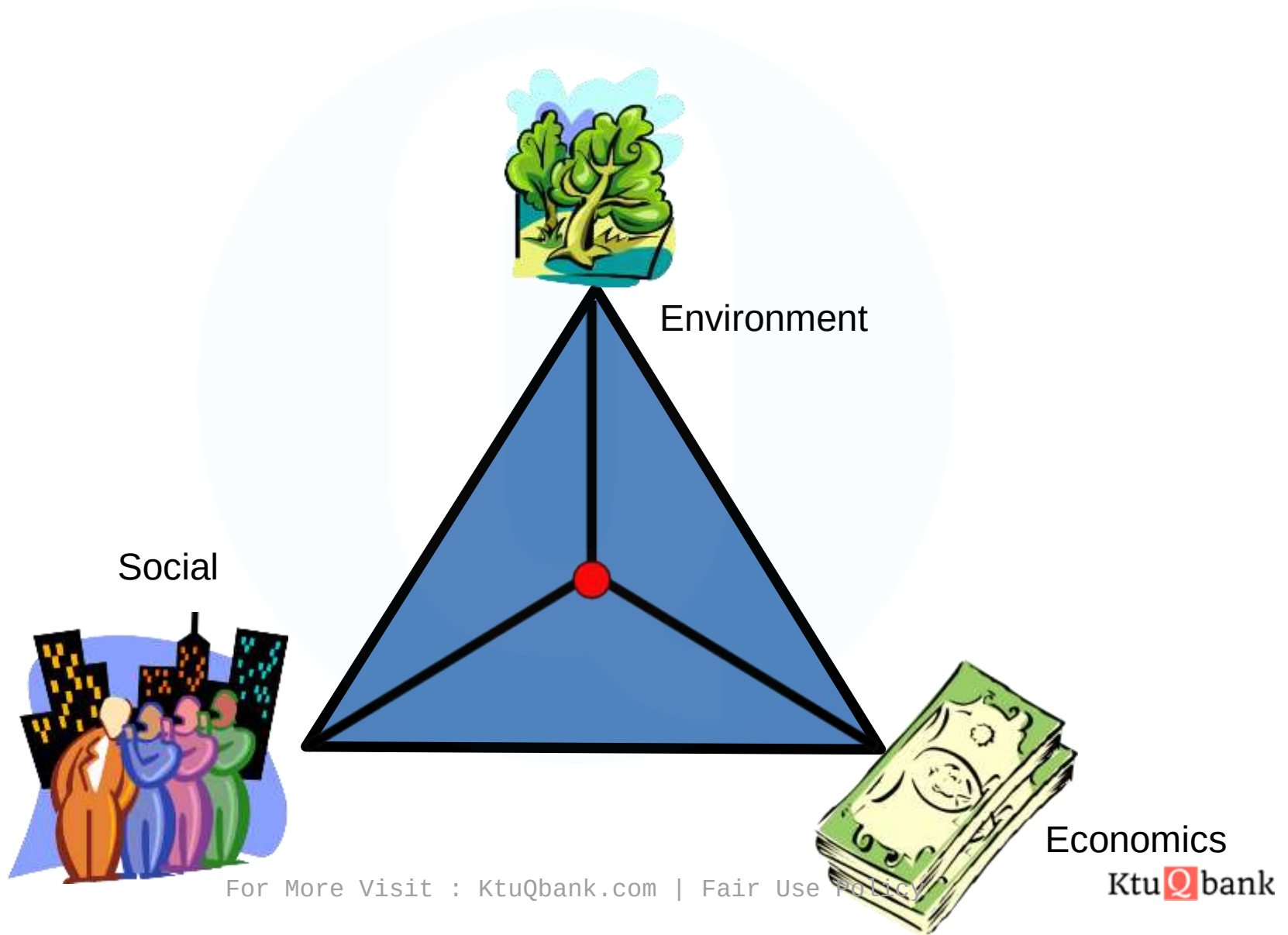


ECONOMIC BENEFITS

- Reduce operating costs
- Create, expand, and shape markets for green product and services
- Improve occupant productivity

SOCIAL BENEFITS

- Enhance occupant comfort and health.
- Improve overall quality of life.





START THINKING GREEN





Sustainable Cities



Inputs

Energy

Food

Water

Raw
materials

Money

Information



Outputs

Solid wastes

Waste heat

Air pollutants

Water pollutants

Greenhouse gases

Wealth

Noise

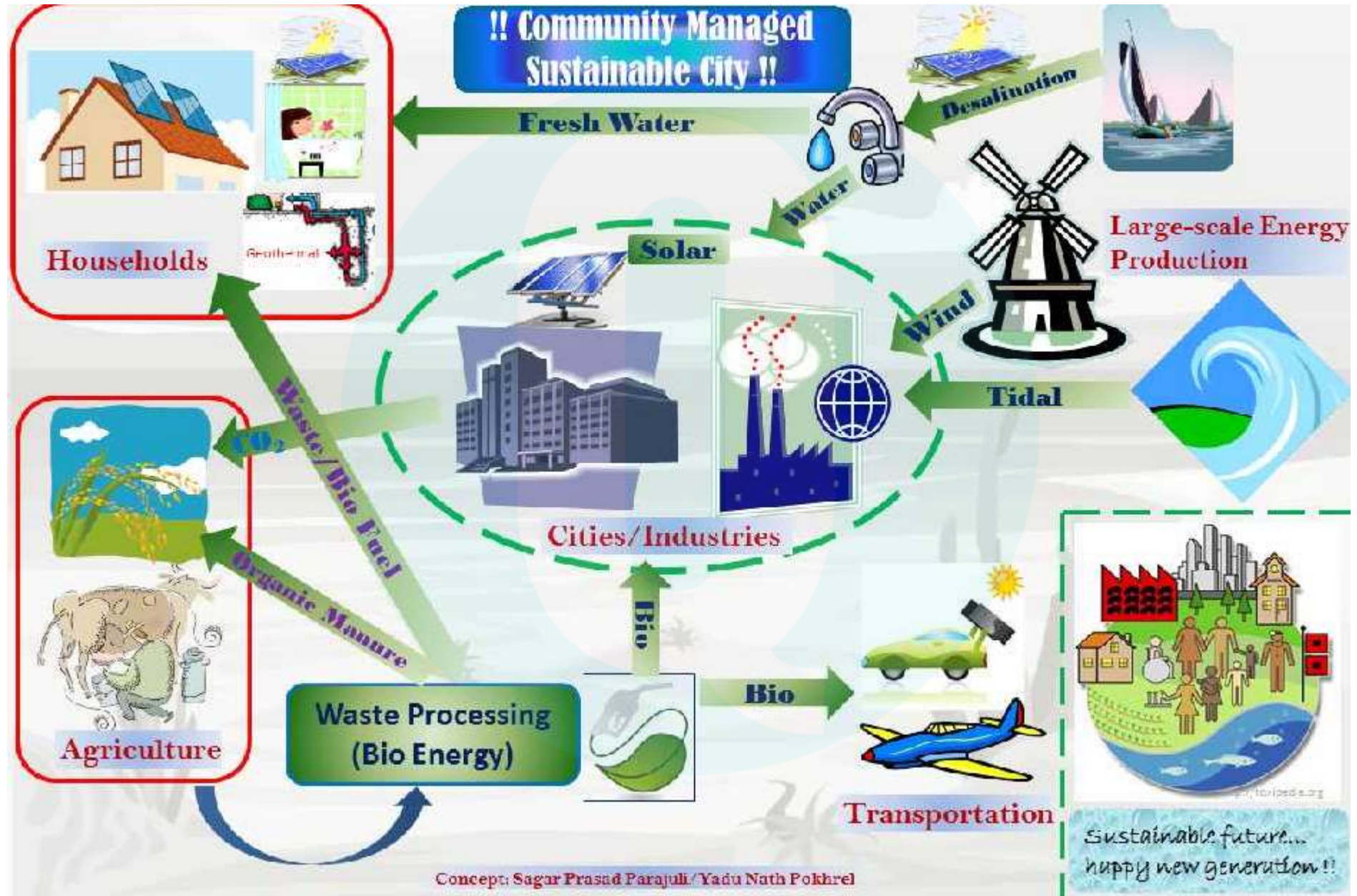
Manufactured goods

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What is a Sustainable City...??



A sustainable city, or eco-city is a city designed with **consideration of environmental impact**, inhabited by people dedicated to minimization of required **inputs of energy, water and food**, and **waste output of heat, air pollution - CO₂, and water pollution**.



Copenhagen

(Sustainable city)





Copenhagen



- The city's airport, rail and suburbs are all connected to the centre by the **metro system**.
- In the city centre, a **combination of measures** has encouraged an **increase in walking and cycling** and a **decrease in private car use**.
- **90 %** of all construction waste is **recycled** and **75 per cent** of all **household garbage** used for heating.
- Many **public squares** and streets are **pedestrianized**.
- The city has a programme to gradually **reduce** the number of **car parking** spaces by **3 per cent** per year and further develop **cycle lanes** and a **free cycle hire** scheme.

Copenhagen goals

- **90% of citizens** will be no more than 15 minutes' walk from a park or one of Copenhagen harbour's two swimming facilities.
- Creation of **green cycle routes** with Government ministers leading by example.
- Currently **37km** out of proposed **110** have been completed



FREIBURG GREEN CITY

Approaches to Sustainability



Eco housing, car-free streets and socially conscious neighbours have made the German city of Freiburg a shining example of sustainability.



- After devastation of WW2, planners rebuilt the city for people **not cars**.
- Pedestrianized streets, **bike lanes** and excellent trams were at the heart of the city's development.
- Plus the unique Freiburg Bächle, small canals that run down each central street.



- **Climate protection** - CO2 emissions by **25 per cent**.
- **Solar Power** - With more than **1,800 hours of sunshine** each year Freiburg is one of the sunniest cities in Germany.
- Freiburg hosts the world's first energy self-sustaining solar building, the **Heliotrope**. Even the local football stadium has become an attraction as the first stadium worldwide to have its own solar plant.
- **Recycling** - The city itself sets a good example by using paper, of which approximately **80 percent** has been recycled.
- **Non-recyclable waste from the region is incinerated at a plant** in the Industrial Park Breisgau, located 20km south of Freiburg. It supplies energy to 25,000 households.

Basic features of a sustainable city

1. Sustainable city should be in harmony with nature
2. More natural open spaces
3. Predominant transport mode to be walking, cycling and public transport
4. Minimise the required inputs such as water, energy, and other resources
5. Conservation of existing water bodies and rain water harvesting
6. Tap more renewable energy sources and reduce relying more on non renewable energy sources
7. Maximise recovery, reuse and recycling of waste materials
8. Maintenance of law and order reduces the crime rates
9. Create jobs for the local population and implementation of poverty elimination programmes
10. Improved waste management

